

## “20 nm allows us to develop heterogenous multi-core SoCs and integrate wider memory in 3D-ICs”

Xilinx, Inc., a programmable logic devices provider, is known for inventing the field-programmable gate array (FPGA) and as the first semiconductor company with a fabless manufacturing model. Founded in Silicon Valley in 1984, the company is headquartered in San Jose, California.

Neeraj Varma, director, sales, Xilinx India, spoke to Ashwin Gopinath of EFY about the company's new portfolio of products in the 28 nm along with the migration into the 20 nm domain

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### What are the things to be kept in mind when designing a big or complex device?

When you design a big device, you are not going to design everything from scratch. You are going to use a lot of IPs. In that case, you either build your own IPs in-house or use the third-party IPs. You need lots of IPs in either case, which is not helped by the fact that most of the design teams today are located far away from each other. What you need is a team-based approach where different people working at different places need to work in tandem.

### What are the advantages of integrating memory on the chip?

Typically, if you look at a board, you have a chip, a processor, FPGA, etc, and then the memory (either a D-RAM or an S-RAM). However, the biggest bottleneck is the bandwidth between memory and chip, which decides how much data you can buffer, process and take out. Now imagine taking the memory inside the chip. A real high-capacity memory inside the chip translates to a higher bandwidth. That is the most significant use of integrating the memory on the chip, which is what we have done on the 20 nm.

Before the second-generation, we never had memory on the chip. Now, by adding memory, we have ensured that the designers don't have to focus on the memory aspect at all. On the transceivers part, earlier we used to do 28 Gbps. Now we are going from 33 to 56Gbps transceivers.

### What are the performance improvements expected from this second-generation portfolio?

From an FPGA perspective, at 20 nm, we are going to see a 30-50 per cent improvement on the price per performance per watt front. What that means is 30-50 per cent price reduction, performance improvement and power reduction, all for the same capability. Our analogue mixed-signal will become mainstream. Right now, we have had a very significant capability at 28 nm. In addition, a lot of IP subsystems will be integrated into it.

From an SoC standpoint, we only had dual-core Cortex A9. Now we are introducing heterogenous multi-core. Plus, the FPGA processing and a software environment that suits it.

For 3D-ICs, it will have the FPGA die. I talked about the transceivers that we have integrated. Now we are also integrating wide memory onto the 3D-IC.